Three Phase AC-DC Converter-II

Introduction

The rapid expansion of electric vehicle (EV) infrastructure necessitates robust and efficient power conversion systems. At the heart of these systems are AC-DC converters, which transform the alternating current from the grid into direct current suitable for battery charging. While single-phase converters serve many applications, high-power charging stations often rely on three-phase AC-DC converters due to their inherent advantages in power delivery and ripple reduction. This lecture delves into the advanced aspects of three-phase AC-DC converters, often referred to as active rectifiers or front-end converters, building upon the foundational understanding of their operation. We will explore the critical design considerations for selecting key components such as inductors, switches, and capacitors, ensuring optimal performance, efficiency, and reliability in demanding charging environments. Understanding these sizing methodologies is paramount for engineers designing the next generation of EV charging solutions.

Core Concepts

Term	Definition	Significance
`Active Rectifier`	A three-phase AC-DC converter that actively controls current flow and power factor, typically using controllable switches like IGBTs or MOSFETs.	Enables bidirectional power flow, unity power factor operation, and precise control over grid current, crucial for grid-friendly charging.
`Modulation Index (M)`	A dimensionless quantity representing the ratio of the modulating signal amplitude to the carrier signal amplitude in PWM schemes.	A designer's choice (typically 0.8-0.9) that influences the voltage applied by the converter and is critical for inductor sizing and voltage drop.
`Unity Power Factor (UPF)`	An operating condition where the current drawn from the AC source is in	Minimizes reactive power, reduces losses, and improves grid efficiency, a key advantage of active rectifiers.

Term	Definition	Significance
	phase with the source voltage.	
`Freewheeling Diode`	A diode connected in parallel with an inductive load or a switch to provide a path for current when the switch opens, preventing voltage spikes.	Essential when using IGBTs, allowing current flow in the reverse direction and protecting the IGBT from overvoltage during switching.
`Harmonic Components`	Frequencies present in a waveform that are integer multiples of the fundamental frequency.	Contribute to current and voltage distortion; active rectifiers aim to minimize lower-order harmonics and manage higher-order ones.
`Second Line Frequency Component`	A harmonic component at twice the line frequency (e.g., 100 Hz for a 50 Hz grid).	A dominant ripple component in single-phase AC-DC converters, but notably absent in the output current of balanced three-phase converters, simplifying capacitor design.
`Pole Voltage`	The voltage at the output of a half-bridge leg, typically referenced to a fictitious midpoint.	Its average variation forms the fundamental frequency component, while its high-frequency variations contribute to harmonic components.
`Parasitic Inductance`	Unintended inductance present in circuit traces, component leads, and wiring.	Causes voltage spikes (L * di/dt) across switches during turn-off, necessitating safety margins in voltage ratings.

Detailed Analysis

1. Inductor Sizing for Current Control

The inductor (L) in a three-phase AC-DC converter plays a crucial role in filtering current and enabling control over power flow. Its value is determined by several factors, including the modulation index (M), peak AC source voltage (Vs,peak), output DC voltage (Vo), line frequency (fs), and peak source current (Is,peak). The lecture highlights that the modulation index is a design choice, typically ranging from 0.8 to 0.9.

A critical design constraint for the inductor is that the voltage drop across it, corresponding to the fundamental frequency component, must be less than 10% of the input peak voltage. This can be expressed as:
`w * L * Is,peak < 0.1 * Vs,peak`
where `w` is the angular line frequency (2 * pi * fs). If this condition is not met, the modulation index (M) may need to be adjusted to achieve the desired L value. The peak source current (Is,peak) can be estimated from the load power (Pl) and phase voltage (Vphase_rms), assuming a lossless converter or incorporating an efficiency term. For a three-phase system, `Is,peak` can be derived from `Pl / (3 * Vphase_rms)` to get RMS current, then multiplied by `sqrt(2)` for the peak.

2. Sizing of Switches: MOSFETs vs. IGBTs

The switches (MOSFETs or IGBTs) are fundamental to the converter's operation, and their proper sizing is essential for reliability. The sizing involves determining the RMS and average current ratings they must withstand.

For MOSFETs with Body Diodes:
In a half-bridge configuration, the duty ratio (D) for the upper switch (S1) is given by `1/2 + m/2 * sin(wt)`, assuming sinusoidal PWM. The lower switch (S2) has a duty ratio of `1 - D`. The RMS current through switch S1 (IRMS_S1) can be calculated by integrating the square of the current over a period, weighted by the duty ratio:
`IRMS_S1 = sqrt(1/T * integral from 0 to T of (D * Is,peak * sin(wt))^2 dt)`
where `Is,peak * sin(wt)` is the assumed unity power factor source current. Solving this integral provides the RMS current rating for the MOSFET.

For IGBTs with Freewheeling Diodes:
When using IGBTs, a freewheeling diode is placed in anti-parallel to allow current flow in the negative direction. Consider a half-bridge with S1 (IGBT) and D1 (freewheeling diode) in the upper leg, and S2 (IGBT) and D2 (freewheeling diode) in the lower leg.
* **Positive Half-Cycle (Vs > 0):**

* When S1 is ON (for DTS period), current flows through S1 (collector to emitter).
* When S1 is OFF (for 1-DTS period), current flows through D2.
* **Negative Half-Cycle (Vs < 0):**
* When S2 is ON (for 1-DTS period), current flows through S2 (collector to emitter).
* When S2 is OFF (for DTS period), current flows through D1.

The average and RMS currents for the IGBT (S1) and the freewheeling diode (D2) can be calculated. For S1, considering only the positive half-cycle where it conducts:

```
I_avg_S1 = (Is,peak / (2 * pi)) * (1 + (pi/4) * m)
```

```
IRMS_S1 = (Is,peak / (2 * sqrt(2))) * sqrt(1 + (8 * m) / (3 * pi))
```

Similarly, for the freewheeling diode D2, which conducts during the 1-DTS period in the positive half-cycle:

```
I_avg_D2 = (Is,peak / (2 * pi)) * (1 - (pi/4) * m)
```

```
IRMS_D2 = (Is,peak / (2 * sqrt(2))) * sqrt(1 - (8 * m) / (3 * pi))
```

These expressions allow designers to select IGBTs and diodes with appropriate current ratings. Similar derivations apply for S2 and D1 by adjusting the integration limits and duty ratios.

3. Voltage Stress Across Switches

Beyond current ratings, switches must also withstand voltage stress. When the lower switch (S2) is ON, the entire output DC voltage (Vo) appears across the upper switch (S1), and vice-versa. Therefore, the voltage rating of S1 and S2 must be at least Vo. However, to account for safety margins and transient effects, the voltage rating is typically chosen as `1.4 * Vo`. This 40% safety margin addresses:
* **Output Voltage Ripple:** The DC output voltage (Vo) is not perfectly smooth and contains some ripple.
* **Parasitic Inductance:** When a switch turns off, the parasitic inductance (L_parasitic) in the circuit loop causes a voltage spike (`L_parasitic * di/dt`) across the switch. This transient voltage adds to the steady-state voltage, necessitating a higher voltage rating.

4. Capacitor Sizing: The Advantage of Three-Phase

The output capacitor (C) filters the DC voltage, and its sizing depends on the ripple current it must handle. In single-phase AC-DC converters, the power balance reveals a dominant second line frequency harmonic component in the output current, leading to significant voltage ripple across the capacitor and requiring a large capacitance.

However, for a balanced three-phase AC-DC converter, the situation is different. The instantaneous input power (Pin) is the sum of the instantaneous powers from each

phase:
`Pin = V_a * I_a + V_b * I_b + V_c * I_c`
Assuming unity power factor and balanced sinusoidal voltages and currents:
`Pin = Vs,peak * Is,peak * [sin^2(wt) + sin^2(wt - 120°) + sin^2(wt - 240°)]`
Using trigonometric identities (`sin^2(x) = (1 - cos(2x))/2`), this simplifies to:
`Pin = (Vs,peak * Is,peak / 2) * [ (1 - cos(2wt)) + (1 - cos(2wt - 240°)) + (1 - cos(2wt - 480°)) ]`
Since `cos(2wt - 480°) = cos(2wt - 120°)` , the terms `cos(2wt) + cos(2wt - 240°) + cos(2wt - 120°)` sum to zero at any instant.
Therefore, `Pin = (3/2) * Vs,peak * Is,peak` .

This derivation shows that the instantaneous input power in a balanced three-phase system is constant and contains **no second line frequency harmonic component**. Consequently, the current flowing into the DC link capacitor (I_converter) also lacks this dominant low-frequency ripple. This is a significant advantage over single-phase converters, as it means the capacitor primarily handles higher-frequency harmonic components originating from the switching action (sidebands of the carrier frequency, mf). These high-frequency components typically have smaller amplitudes, allowing for a much smaller output capacitor size compared to three independent single-phase converters. The exact calculation of the RMS current through the capacitor due to these high-frequency harmonics will be further explored in subsequent discussions.

Key Takeaways

- [ ] Three-phase AC-DC converters (active rectifiers) enable bidirectional power flow and unity power factor operation.
- [ ] Inductor sizing depends on modulation index, source/output voltages, and line frequency, with a critical constraint on voltage drop (less than 10% of Vs,peak).
- [ ] Switch sizing requires calculating RMS and average currents, which differ for MOSFETs and IGBTs with freewheeling diodes.
- [ ] Duty ratios for switches are derived from sinusoidal PWM, influencing current conduction periods.
- [ ] Voltage stress on switches must account for output DC voltage, ripple, and parasitic inductance effects (typically 1.4 times Vo).
- [ ] Balanced three-phase converters inherently eliminate the second line frequency harmonic component in the output current.
- [ ] The absence of the second line frequency harmonic allows for a significantly smaller output capacitor compared to single-phase systems.

- [ ] Output capacitors in three-phase converters primarily filter high-frequency harmonic components from switching.

Conclusion

The detailed analysis of component sizing for three-phase AC-DC converters underscores the sophisticated engineering required to achieve high performance in modern power electronics. From precisely calculating inductor values to ensure proper current control and voltage drop, to meticulously rating switches for both current and voltage stresses, every component plays a vital role. The inherent advantage of three-phase systems in eliminating the problematic second line frequency harmonic component in the output current is a cornerstone of their efficiency and compactness, allowing for smaller, more cost-effective capacitor banks. This understanding is crucial for designing robust and grid-friendly charging infrastructure, where converters must not only deliver power efficiently but also interact harmoniously with the electrical grid. Future discussions will further refine the capacitor sizing methodology by quantifying the high-frequency ripple currents, completing the comprehensive design picture for these essential power conversion units.